



The Scots College

Year 12

HSC Extension 1 Mathematics

Task 1

February 2013

Time Allowed: 45 minutes

Weighting: 10%

Instructions to Students:

- Show all working
- Board approved calculators may be used
- Table of Standard Integrals attached
- Start a new booklet for each Question.

Question/ Learning Intention	Polynomials	Applications of Calculus
1	/9	
2	/7	

3		/10
4		/11
Total		
/37		

Question 3: 10 Marks (start a new booklet)

Marks

a) Find $\int x\sqrt{9-x^2} \, dx$, using the substitution $u = 9 - x^2$. **3**

b) Find $\int \frac{1-x}{(1+x)^3} \, dx$, using the substitution $u = 1 + x$. **3**

c) At any point on the curve $y = f(x)$ the gradient function is given by $\frac{dy}{dx} = 2\cos^2 x + 1$. If $y = \pi$ when $x = \pi$, find the value of y when $x = 2\pi$. **4**

Question 4: 11 Marks (start a new booklet)

2

a) Evaluate $\int_0^\pi 2\sin x \cos^2 x \, dx$

b) The area bounded by the curve $y = \sin x$ between $x = 0$ and $\frac{\pi}{2}$ and the x-axis is rotated about the x-axis. Find the volume of the solid of revolution. **3**

c) i) By expanding $\cos(2\theta + 2\theta)$ or otherwise, show that : 3

$$\cos 4\theta = 1 - 8\sin^2\theta + 8\sin^4\theta$$

ii) Hence evaluate $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\sin^2 \theta - \sin^4 \theta) d\theta$ 3

END OF EXAM

Question 1 9 Marks (start a new booklet) **Marks**

a)i) Given the polynomial $P(x) = 2x^3 - 9x^2 + kx + 6$, find the value of k if $(x - 3)$ is a factor of $P(x)$. 2

ii) Hence, or otherwise, determine all the roots of the equation $P(x) = 0$. 3

b) If the equation $5x^3 - 6x^2 - 29x + 6 = 0$ has roots α, β, γ find the value of the following (leave your answer in fraction form) :

i) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ 2

ii) $\alpha^2 + \beta^2 + \gamma^2$ 2

Question 2 7 Marks (start a new booklet)**Marks**

- a)i)** Show that the equation $f(x) = x^3 - 8x + 8$ has a zero between $x = -3$ and $x = -4$. **2**
- ii)** Taking $x = -3.5$ as the first approximation of the solution of the equation $f(x) = 0$, use Newton's method once to find a closer approximation, giving your answer to 2 decimal places. **2**
- b)** The polynomial $P(x) = 6x^3 - 7x^2 + ax + b$ has a zero at $x = -1$ and the remaining zeros are reciprocals $(\alpha, \frac{1}{\alpha})$. Determine the value of a and b . **3**

* Record Results

on top of each
Section

Yr 12 Mathematics Extension 1

SOLUTIONS: Task 1. Feb 2013

- 2.25 p
- ① Peter
 - ② Leonard
 - ③ Dave
 - ④ Arun

Q 1

(a) (i) $P(x) = 2x^3 - 9x^2 + kx + 6$

$P(3) = 0$

$2(3)^3 - 9(3)^2 + 3k + 6 = 0$ ✓

$54 - 81 + 3k + 6 = 0$

$3k = 21$ ✓

$k = 7$

(ii)

$$\begin{array}{r} 2x^2 - 3x - 2 \\ x-3 \overline{) 2x^3 - 9x^2 + 7k + 6} \\ \underline{2x^3 - 6x^2} \\ -3x^2 + 7x \\ \underline{-3x^2 + 9x} \\ -2x + 6 \\ \underline{-2x + 6} \\ 0 \end{array}$$
 ✓

$2x^2 - 3x - 2 = (2x + 1)(x - 2)$

$\therefore P(x) = (x-3)(x-2)(2x+1) = 0$ ✓

$\therefore x = \underline{3, 2 \text{ or } -\frac{1}{2}}$ ✓

2.28

Q 1
(b)

$$5x^3 - 6x^2 - 29x + 6 = 0$$

α, β, γ are roots.

$$\alpha + \beta + \gamma = 6/5$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = -29/5$$

$$\alpha\beta\gamma = -6/5$$

$$(i) \quad \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} \quad \checkmark$$

$$= \frac{-29/5}{-6/5} \left(\frac{1}{2}\right)$$

$$= \underline{\underline{29/6}} \left(\frac{1}{2}\right) \checkmark$$

(2)

$$(ii) \quad \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) \quad \checkmark$$

$$= \left(\frac{6}{5}\right)^2 - 2\left(-\frac{29}{5}\right)$$

$$= \frac{36}{25} + \frac{58}{5}$$

$$= \frac{36 + 290}{25}$$

$$= \underline{\underline{\frac{326}{25}}} \quad \checkmark$$

(note : question asked for fraction form)

Question 2

(a) (i) $f(x) = x^3 - 8x + 8$

$$\begin{aligned} f(-3) &= (-3)^3 - 8(-3) + 8 \\ &= -27 + 24 + 8 \quad \left(\frac{1}{2}\right) \\ &= 5 > 0 \end{aligned}$$

$\frac{1}{2}$ each sub.

1 for explanation

$$\begin{aligned} f(-4) &= (-4)^3 - 8(-4) + 8 \\ &= -64 + 32 + 8 \quad \left(\frac{1}{2}\right) \\ &= -24 < 0 \end{aligned}$$

total.
2

(1) ← reasoning

∴ A root lies between $x = -3$ and $x = -4$.

(ii) $x = -3.5$ $f(x) = x^3 - 8x + 8$

$$\begin{aligned} f(-3.5) &= -6.875 & f'(x) &= 3x^2 - 8 \\ f'(-3.5) &= 28.75 \end{aligned}$$

$$x_1 = x - \frac{f(x)}{f'(x)}$$

$$= -3.5 - \frac{-6.875}{28.75} \quad \checkmark$$

$$= -3.2608 \dots$$

$$\approx \underline{\underline{-3.26}} \quad \checkmark$$

Q2(b) $P(x) = 6x^3 - 7x^2 + ax + b$

$$P(-1) = 0.$$

The roots are $-1, \alpha, \frac{1}{\alpha}$.

$$\text{Prod of roots} = -1 \times \alpha \times \frac{1}{\alpha} = -1 = -\frac{b}{6}.$$

$$\therefore \boxed{b = 6} \quad \checkmark$$

$$-1 + \alpha + \frac{1}{\alpha} = \frac{7}{6}.$$

$$\text{or } \alpha + \frac{1}{\alpha} = \frac{7}{6} + 1$$

$$\alpha + \frac{1}{\alpha} = \frac{13}{6} \quad \text{--- (1)}$$

} Sum roots
 $\checkmark$

~~Prod~~ Sum roots 2 at a time.

$$-1 \times \alpha + -1 \times \frac{1}{\alpha} + \alpha \times \frac{1}{\alpha} = \frac{a}{6}$$

$$\text{or } -\alpha - \frac{1}{\alpha} + 1 = \frac{a}{6},$$

$$\text{or } -(\alpha + \frac{1}{\alpha}) + 1 = \frac{a}{6}$$

$$\text{or } -\frac{13}{6} + 1 = \frac{a}{6} \quad (\text{from (1) } \alpha + \frac{1}{\alpha} = \frac{13}{6})$$

$$\text{or } \frac{a}{6} = -\frac{7}{6}.$$

$$\therefore \boxed{a = -7} \quad \checkmark$$

$$\therefore \underline{\underline{a = -7, b = 6}}$$

Q 3

$$(a) I = \int x \sqrt{9-x^2} dx \quad u = 9-x^2$$

$$\frac{du}{dx} = -2x$$

$$\frac{du}{-2} = x dx$$

$$I = \int u^{\frac{1}{2}} \left(\frac{du}{-2} \right)$$

$$= -\frac{1}{2} \int u^{\frac{1}{2}} du$$

$$= -\frac{1}{2} \left[\frac{u^{\frac{3}{2}}}{\frac{3}{2}} \right] + C$$

$$= -\frac{1}{3} (9-x^2)^{\frac{3}{2}} + C$$

$$(b) \int \frac{1-x}{(1+x)^3} dx$$

$$u = 1+x$$

$$du = dx$$

$$= \int \frac{2-u}{u^3} du$$

$$x = u-1$$

$$1-x = 1-u+1 = 2-u$$

$$= \int (2u^{-3} - u^{-2}) du$$

$$= \frac{2u^{-2}}{-2} - \frac{u^{-1}}{-1} + C$$

$$= -\frac{1}{(1+x)^2} + \frac{1}{1+x} + C$$

Q4

(a)

$$\int_0^{\pi} 2 \sin x \cos^2 x \, dx$$

$$\cos x = u$$

$$-\sin x \, dx = du$$

$$\sin x \, dx = -du$$

$$= 2 \int_{-1}^1 u^2 (-du)$$

$$x=0, u=1$$

$$x=\pi, u=-1$$

$$= -2 \int_{-1}^1 u^2 \, du$$

$$= -2 \left[\frac{u^3}{3} \right]_{-1}^1$$

$$= -2 \left[\frac{1}{3} - \frac{-1}{3} \right]$$

$$= -2 \left(-\frac{2}{3} \right)$$

$$= \frac{4}{3}$$

2.42

Question 3

(c) $\frac{dy}{dx} = 2\cos^2 x + 1$ (π, π) is a point.

$$y = \int (2\cos^2 x + 1) \, dx$$

$$= \int (1 + \cos 2x + 1) \, dx$$

$$= \left[2x + \frac{1}{2} \sin 2x \right] + C$$

Sub (π, π)

$$\pi = 2\pi + \frac{1}{2} \sin 2\pi + C$$

$$C = -\pi$$

$$y = 2x + \frac{1}{2} \sin 2x - \pi$$

When $x = 2\pi$

$$y = 4\pi + \frac{1}{2} \sin 4\pi - \pi = 3\pi$$

2.44

Question 4 b)

$$V = \pi \int_0^{\pi/2} y^2 dx$$

$$= \pi \int_0^{\pi/2} \sin^2 x dx$$

$$= \frac{\pi}{2} \int_0^{\pi/2} (1 - \cos 2x) dx$$

$$= \frac{\pi}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\pi/2}$$

$$= \frac{\pi}{2} \left[\left(\frac{\pi}{2} - \frac{1}{2} \sin \pi \right) - \left(0 - \frac{1}{2} \sin 0 \right) \right]$$

$$= \frac{\pi}{2} \cdot \frac{\pi}{2}$$

$$= \frac{\pi^2}{4} \underline{\underline{u.s.}}$$

2.45

Question 4

$$\cos(2\theta + 2\theta)$$

C) (i)

$$= \cos^2 2\theta - \sin^2 2\theta.$$

$$= 1 - 2\sin^2 2\theta - 4\sin^2 \theta (\cos^2 \theta).$$

$$= 1 - 4\sin^2 \theta + 4\sin^4 \theta - 4\sin^2 \theta (1 - \sin^2 \theta)$$

$$= 1 - 4\sin^2 \theta + 4\sin^4 \theta - 4\sin^2 \theta + 4\sin^4 \theta.$$

$$\cos 4\theta = 1 - 8\sin^2 \theta + 8\sin^4 \theta.$$

$$(ii) \quad 1 - 8\sin^2 \theta + 8\sin^4 \theta = \cos 4\theta.$$

$$8\sin^2 \theta - 8\sin^4 \theta = 1 - \cos 4\theta.$$

$$\therefore \sin^2 \theta - \sin^4 \theta = \frac{1}{8} (1 - \cos 4\theta)$$

$$\int_{\pi/4}^{\pi/2} (\sin^2 \theta - \sin^4 \theta) d\theta = \frac{1}{8} \int_{\pi/4}^{\pi/2} (1 - \cos 4\theta) d\theta.$$

$$= \frac{1}{8} \left[\theta - \frac{1}{4} \sin 4\theta \right]_{\pi/4}^{\pi/2}$$

$$= \frac{1}{8} \left[\left(\frac{\pi}{2} - \frac{1}{4} \sin 2\pi \right) - \left(\frac{\pi}{4} - \frac{1}{4} \sin \pi \right) \right]$$

$$= \frac{1}{8} \left(\frac{\pi}{2} - \frac{\pi}{4} \right)$$

$$= \frac{\pi}{32}$$